

Cardiovascular system
Case sheet format.

DATE: 11/3/2019

Name
Age
Sex
Occupation
Address

Complaints & duration

Symptoms for which the patient has come to hospital
&
Symptoms told by patient after questioning
in chronological order.

H/O presenting illness:

The patient was apparently normal before _____
after which he developed,

H/O Breathlessness/dyspnea

Onset

duration

NYHA class

Orthopnea - present/Absent, Severity

Paroxysmal Nocturnal dyspnea - Present/Absent.

Symptoms told as
complaints by patients
to be elaborated 1st
then other symptoms

PND

Mechanism

① Reabsorption of fluid
from tissues into plasma

② ↑ sed venous return to heart

③ Fluid redistributed into lungs

④ ↑ in LA pressure & fall in PaO_2 during sleep.

Features

- Sudden dyspnea - wake
pt from sleep

- gasp for air, open window

- cough, frothy sputum

- pallor

- sweating

- tachycardia

- crackles

REDMI NOTE 6 PRO
MI DUAL CAMERA

Young age → congenital cyanotic heart disease, RHF → 5-15 yrs age
Middle age → Rheumatic fever sequelae, Ischaemic heart disease, CAD
Old age → Heart failure

Men → develop CAD earlier
Women → > 65 yrs Risk of CAD M=F

Diabetic women → Symptoms - very subtle. → mild heart burn
 But in Men → Symptoms - typical
 (People without diabetes) chest pain

CARDINAL SYMPTOMS

- dyspnoea
- chest pain
- palpitation
- syncope
- cough with expectoration
- oedema
- fatigue
- fever

Hypnoea - dyspnoea on lying flat within 10 minutes
 ↓
 Advanced heart failure (CHF)
 or
 Incipient decompensation

flat → ↑ venous return
 LV dysfunction → pulmonary edema

Severity → 1 pillow orthopnoea
 2 pillow " "
 3 pillow " "

Dyspnoea

Acute
 - Heart failure
 - pulmonary embolism
 - Arrhythmias

Acute heart failure

prefer to be upright.
 Abb - pH with pulmonary edema

Chronic

- Heart failure (decompensated)
 - cardiogenic edema

Chronic heart failure

Exertional dyspnoea
 HALLMARK

Pulmonary embolism

more comfortable lying flat.
 If sat upright → syncope

Platypnoea (sitting/standing)

Triepnoea (lateral recumbent)

Venular thrombosis
 PFA → R to L shunting
 chronic heart failure
 pt. avoided
 lateral recumb position

Only NYHA grade IV dyspnoea → orthopnoea + +

Dyspnea - Abnormal awareness of one's own breathing disproportionate to the degree of exertion

NYHA

Physical activity		Cause
I	No limitation Ordinary physical activity & no undue PDPF	Asymptomatic LV dysfunction
II	Slight limitation Ordinary physical activity ↳ PDPF, Angina comfortable @ rest	symptomatic mild HF
III	Marked limitation Less than ordinary activity	symptomatically mod HF
IV	Severe PDPF @ rest ↑↑ discomfort with any	symptomatically severe HF

PND - Earliest symptom of LHF.

↓ occurs within 2-4 hrs of lying flat during sleep

LV tries to pump out blood in this (compensate) 2-4 hrs but soon fatigues

↓
fluid into lungs
↓
Pulmonary congestion.

File
Date: / /

H/O chest pain

Site

Onset

Character

Radiating

Associated

Timing

Exacerbating/relieving factors

Severity

Angina - "well described
tight / pressing / band like"
due to heavy weight.

Stable Angina -
- lasts < 5 min
- occurs at predictable times
- relieved by rest

Unstable Angina -
- Sudden severe
- lasts > 20 min
- occurs at unpredictable times
- not relieved by rest

Myocardial Infarction -
- Lasts > 20 min
- severe pain
- not relieved by rest

Associated features

breathlessness

N, V, breathlessness

sweating

feeling of impending death
[Anger animi]

syncope, sweating
focal neurological signs
limb ischaemia
miserere cruraemia

breathlessness, Fever
flu-like prodrome

Heart burn,
Acid reflux

Cause

Coronary atherosclerosis
AS, Myocarditis,
Coronary spasm

plaque rupture
CA - Occlusion

Character

MI - Thoraic aortic
dissection rupture

Ras

Timing

Acute presentation

prolonged duration - MI, AD

Variable duration - pericardial pain

Severity

Mild to moderate - Angina

Severe - MI, Pericardial

Very severe - AD

Overexposed pain

Deep chest
spasm, reflux
intercostal

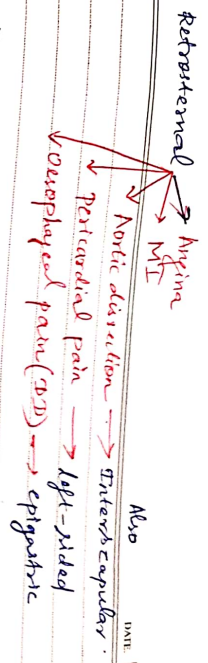
Intermittent

eps lasting 2-10 mins - Angina

@ night time / variable duration - Overexposed pain

Mild but overexposed - Overexposed pain
mimics MI

Site



Onset

Very sudden → Aortic dissection
Rapid over few mins → MI
Progressive increase over 1-2 mins → Angina
Gradual → pericardial pain
Over 1-2 mins / sudden → Oesophageal pain

Character

Constricting, heavy → Angina, MI
tearing/ripping → Aortic dissection (AD)
Sharp/stabbing/pleuritic → pericardial pain
gripping, tight/burning → oesophageal pain

Radiation

Arm, neck, epigastrium → Angina

Arm, neck, epigastrium → MI
↓ jaw
Usually spontaneous
Rapid ③ - stress, exercise
Not ④ - rest, nitrates

Back b/w shoulders → AD
Spontaneous
Noting relieve pain

Left shoulder/back → pericardial pain ③ - sitting up/lying
④ - NSAIDs

Back / → Oesophageal pain ③ - lying flat, spicy food
Not ④ - rest, nitrate

Exacerbating/relieving factors

W/O palpitation could -

Weakness character -

Tiring

Triggers

Relieving factors

Associated symptoms

Extrasystoles → Benign

Ectopic beats

Missed beat



Premature ectopic beat

→ - small stroke volume due to
incomplete LV filling
↓
Impaired pulse

felt by

strong heart beat



compensation

by LV overfilling

(Jump/Thump)

& forceful contraction

with next beat

Physiological	Pathological
Exercise	Arrhythmia
Anxiety	Hyperthyroidism
Pregnancy	MI, A2, T2 (Volume overload) in the cluster irritating about conduction of leading to arrhythmia

PALPITATION

Unpleasant awareness of one's own heart beating in the chest

Healthy people

→ Normal sinus rhythm
↓
awareness after exercise
or
Anxiety

More common @ night

Sleep people → when lying on left side.

OTC

Onset

Sudden

→ Extrasystoles
→ Supraventricular tachycardia (with jump)
→ AF
→ Ventricular tachycardia
→ Sinus tachycardia

Gradual

→ Sinus tachycardia

Timing

Timing

Timing

Timing

Timing

Character

Regular, fast

→ Sinus tachycardia (Pounding) ST
→ Supraventricular tachycardia SVT
→ Ventricular tachycardia VT

Irregular < fast
< slow
(elderly) → AF

Jump; missed beat / flutter → Extrasystoles

Exacerbating / relieving factors

→ Extrasystole

③ Fatigue, caffeine, alcohol

③ Walking → ↑ sinus rate

③ Rest, forced movements (bending)

③ Vagal manoeuvres

③ Often spontaneous

③ Exercise / alcohol

③ Exercise

Associated factors

→ Polymia, light-headedness
chest tightness

→ AF → polyuria, breathlessness, syncope

→ VT → presyncope, syncope, chest tightness

→ ST → Anxiety

1/10 syncope.

episodes/day

duration of each episode -

Any Triggers - exertion/intense emotion/prolonged standing/postural change.

Any preceding symptoms
(if syncope)

chest pain
palpitation
lightheadedness
nausea, sweating
visual disturbance
tinnitus

Any associated symptoms
(if presyncope)

gagging

Recovery time to full consciousness -

Vasovagal syncope → Abnormal autonomic reflex → Sudden bradycardia

Vasodilatation

Triggers - prolonged standing
intense emotion

pain

Prodrome - Lightheadedness,

Nausea, sweating
visual disturbance
(dimming from periphery)
tinnitus

Duration - < 60 sec

Wz lying flat after losing consciousness → ↑ cerebral circulation

If person held upright → duration ↑

↓
↑ cerebral hypoperfusion

Recovery - rapid

no confusion

↓
Seizures.

Types of syncope

Neurologically mediated - Vasovagal
Situational - cough, micturition

Cardiac arrhythmias

Orthostatic - postural hypotension

Cardiac arrhythmias

structural, - AS

HOCM

Syncope

Presyncope → sense of lightheadedness & impending loss of consciousness without progressing to actual LOC.

Transient loss of consciousness due to transient cerebral hypoperfusion usually associated with loss of postural tone. (10-30 sec)

Causes

- Postural hypotension
- Neurocardiogenic syncope (Vasovagal)
- Arrhythmias
- Medr obstruction to cardiac O/P. → AS HOCM aortic dissection, mitral stenosis, coronation of aorta.
- Massive pulmonary embolism

Postural hypotension

Fall of > 20 mmHg in sys BP on standing.

3rd Old people > 65 yrs

hypovolemia

Autonomic neuropathy

↓ autonomic failure
↓ autonomic failure (DM, Parkinson's)

Drugs → Vasodilators ACS inhibitors, ARBs, nitrates, & blockers
(anti-hypertensives) (Heart limiting agents) - β-blockers, Ca channel blockers, digoxin

Arrhythmias

Bradyarrhythmias - caused by SA disease / AV block (Stokes-Adams attacks)

Tachyarrhythmias

Ventricular - causes syncope esp. in LV dysfunction
Supraventricular - (AF) rarely causes syncope.

Long QT interval

Mech. obstruction to cardiac O/P - AS, HOCM → esp. on exertion due to increased metabolic demand

Massive pulmonary embolism → obstructed outflow from RV - ASD with acute dyspnea, hypoxia, chest pain

Rare causes

Atrial myxoma
Dissecting aortic aneurysm - thrombosis/failure

H/O Cough with expectoration

onset

quantity

colour

odour

blood-stained / not

postural
diurnal variation
aggravating / relieving factors

H/O edema

location

unilateral / bilateral / anasarca

onset

type - pitting / non-pitting

H/O fatigue - duration
NYHA grade

H/O fever

onset

progression, continuous / intermittent

diurnal variation

Associated features -> chills / rigor / night sweats

cough with expectoration

fixed

Chronic pulmonary congestion
due to
Mitral regurgitation disease

Neuropathic → are pulmonary capillaries rupture
Bronchitis → blood stained mucus with heart failure cells

Edema

Excess fluid in interstitial space.

Anasarca

Nephrotic syndrome
Cardiac failure
Liver cirrhosis
Hypoproteinaemia

Unilateral

DVT
Superficial vein
harvested for CABG

Bilateral

Heart failure
Chronic venous disease
Ca²⁺ channel blockers (nifedipine)
Hypoproteinaemia

↑ se in SVP → strongly suggest cardiogenic cause of edema.

Location

Pedal → Abdomen → Face (periorbital)

Face → Abdomen → Legs → Abdomen

Abdomen → Legs → Face (Ascites)

Cardiac

Renal

Hepatic

Fever

Causes

Infective endocarditis

Pericarditis

Acute MI

Rheumatic fever

Rheumatic reaction

Atypical myxoma (rare).

Advanced heart failure

Prolonged weight loss
Catabolic state
Muscle wasting → Cardiac cachexia

Abdominal distension (Ascites)

1/10 similar complaints in the past

1/10 suggestive of rheumatic fever

1/10 DM, HT, dyslipidaemia, connective tissue disorders

Raynaud's
pericarditis

1/10 skin infection, penetrating trauma, recent dental work

1/10 hospitalisation & surgeries

CAD
↓
IHD

RF

IHD { Previous MI / unstable angina

Try endocarditis

Valvular disease

Pericardiac coronary intervention

(angioplasty, stenting)

CABG

IHD

Transcatheter Cardioverter-defibrillator

cardiac resynchronisation therapy

HF

Surgical valve repair / replacement

Transcatheter valve procedures

Valvular disease

1/10 any major investigations

Aspirin, NSAIDs, Thiazine → Angina. TAN

β-blockers (Asthma) } → Dyspnoea RLN
Ca channel blockers }
 NSAIDs (Verapamil, diltiazem)

β₂ agonists (salbutamol)
Thyroxine
IC_A
 digoxin toxicity
 diuretics → hypokalaemia } → palpitation RLN

Chemotherapeutic
Toxinosis
 dilated
 cardiomyopathy

Anti-hypertensive vasodilators

POSTURAL
HYPOTENSION

ACE Inhibitors
 ARBs
 α blockers
 nitrates

Syncope /
 presyncope
 ↓
 Enalapril
 Cardiac remodeling

CARDIAC
ARRHYTHMIAS
BRADY ARREST

β blockers
Ca channel blockers
 digoxin (Verapamil, diltiazem)

Calcium antagonists

NSAIDs

Ca channel blockers
 (Amlodipine)
 (Nifedipine)

→ Edema

H/O sudden unexplained death @ young age.

Cardiomyopathy
inherited arrhythmia.

PERSONAL HISTORY

Diet - Veg / non Veg.

Thiamine deficiency → wet ber-beri → Cardiac failure.
Selenium deficiency → Keshan disease → Myocardial necrosis.

Alcohol

_____ yrs

→ (Holiday heart syndrome)
Binge drinking → AF

quantity _____ (g)

Independent } HT, obesity
risk factor

Type

Dilated cardiomyopathy

GRE

OCCUPATIONAL HISTORY

DATE

Heart disease → causes physical limitation

↓
Affects employment

↓
Anxiety

Occupational vibration exposure → hand-arm vibration syndrome

↓
Vibration - symptoms

↓
Raynaud's

↓
Numbness

↓
tingling

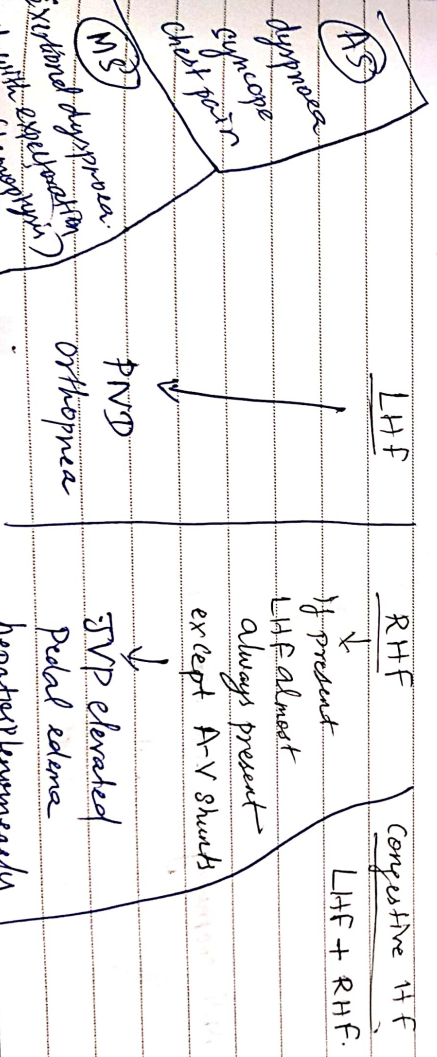
If diagnosis of IHD arrhythmia → in pilots, drivers

↓
can't continue flying

↓
for public safety

Summary

Probable system involved → CVS



MI DUAR CAMERA
6 PRO

Conscious
Oriented
Comfortable → not comfortable but not
Cooperative
built
Nausea/blood? → poor → if some bile has
cardiac contents [THF]
(K+ F) → liver enzymes
↓
hard nutrient absorption
↓
bad appetite

H/O suggestive of - Anemia

Jaundice
cyanosis
clotting
pedal edema - Uni/Bi
pitting/non pitting
gen. lymphadenopathy

Markers / → Rheumatic activity

Wigand's polyarthritis
EC nodules, syphilis
syndromal chorea

Evidence of

Infective endocarditis

Tendon xanthomas
Xanthelasma
corneal arcus
Early Atherosclerosis

congenital heart disease
(Frank's sign)
↓
Down's
Turner's
Marfan's
Pectus excavatus
on conjunctiva
ROH is 2 spots X

Anthropometry → HT → Tall / short stature
wt → obese?

BMC

new
to occlude the

Accessibility of peripheral pulses

BP - 2 UL + 1 LL $\rightarrow$ 120/80 mm Hg^{avg}, Right Arm, Supine

Temp. —

RR —

JVP — Right atrial pressure $\rightarrow$ Central Venous Pressure.

① $N \leq 9$ cm. Sterna angle 5 cm above right atrium +

JVP — cm vertically above sternal angle @ 45° ① $N \leq 4$

RAP Low $\rightarrow$ Lie flat

RAP high $\rightarrow$ sit upright } for JVP to be seen.

2017

2017 guidelines for HT

Primary HT

<120 / <80 → 0 BP

Secondary HT

120-129 / <80 → elevated BP

Stage I HT

130-139 / 80-89 → Stage I HT

Stage II HT

≥140 / ≥90 → Stage II HT

Many asymptomatic may have headable & weight for exercise

pulse → pressure wave generated by ejection of blood into circulation from LV

Diff. of >10mmHg b/w 2 arms

aortic / outflow artery disease.

Cuff size → 80% of length / upper

40% of width / circumference

Obese pts → std adult cuff → Overestimate BP

Why palpation method done 1st?

To not miss the 1st Korotkoff sound

else sys. pressure will be recorded at a falsely low level.

20% elderly HT pts →

Korotkoff sounds appear @ systolic BP.

2 disappear in the interval b/w

sys & diastolic BP.

PATIENT'S ELBOW - LVL WITH HEART

arm elevated by 7cm → 5mm Hg change in

Sys & diast BP

Nothing

End organ damage

Heart failure

CAD, peripheral

stroke

renal

diabetic

hypertensive

from LV

that circulating blood exerts against arterial wall.

Systolic pressure → Max. pressure - during systole (ventricular contraction)

Diastolic pressure → Pressure - maintained @ low level by elasticity & compliance of vessel wall during diastole (ventricular filling)

Korotkoff sounds → Produced b/w systolic & diastolic B.P.
Artery collapses completely & reopens with each beat

↓
Snapping / knocking sound.

1st Thud 120 mmHg systolic

2nd Blowing noise 110 mmHg

3rd Softer thud 100 mmHg

4th Disappearing blowing noise 90 mmHg diastolic (1st)

5th Nothing 80 mmHg diastolic (2nd)

Nothing

③ Abdomenogram

↓
Pres JVP.
LIDS.

④ Position

(Supine)

jugular vein distended - pulsation not seen

(@ 45°)

pt b/w distended & collapsed vein

↓
just above clavicle.

(Upright)

jugular vein - upper part collapsed.

Q₁ - 1 come together

Cannon waves → Irregular → complete heart block (ac waves)
Regular → junctional rhythm
VT
SVT

Elevated JVP → RHF. → + Sustained abdominojugular
reflex > 10s

Pulmonary embolism
chronic pulmonary HT
Cardiac tamponade

+ prominent y descent ← Pericardial constriction
Pericardial effusion

+ loss of pulsation ← SVC obstruction
by lung cancer
no longer reflects RAP
Abdominojugular test -ve

Kussmaul's sign
Paradoxical ↑ JVP
on inspiration
Severe RHF
Pericardial
constriction
Restrictive cardiomyopathy

SVT

If JVP elevated too much
so that it can't be seen (goes very up)

↓

Use manometer
for measurement.

REDMI NOTE 6 PRO
MI DUAL CAMERA

Systemic examination

CVS

Inspection

Chest wall Symmetry

Pectus excavatum / (Funnel chest) (Pigeon chest) → affect heart position
 (Cobbler's chest)
 Kyphosis / scoliosis / kyphoscoliosis

Precordial bulge

Traces - midline / not

only normal ← Apical impulse - location

abnormal ← other pulsations

Suprasternal

supracardiac

parasternal

epigastric

Pulsations above heart

always

Aortic valve disease

Surgical scars

sinuses

Left Submammary Scar

Infracardiac scars

- Pacemaker
 - defibrillator

Midline sternotomy

- CABG
 - Valve replacement

- Mitral valve

Transapical

transcatheter

aortic valve replacement

implantation

bulge - obvious

Stand behind the patient (6 ft) - b/w doc & pt.

Inspect

Level of both shoulders
Inferior angle of scapula on both sides
Inferior angle of scapula to mid vertebra equal in distance?
For chest symmetry

Kyphosis - (Lateral side)

Scoliosis - 'S' shaped

Palpation

Trachea - confirmed to be in midline

Apical impulse - Location → (N) left 5th ICS

Character

Thrill

Parasternal pulsation

(Mid pt b/w suprasternal notch & acromioclavicular joint)

Light touch ← Palpable impulse ← Heave
tactile equivalent ← Palpable vibration ← Thrill

palpable P₂ → keep finger over left 2nd ICS

Left parasternal heave

↓
R Ventricular hypertrophy dilatation

↓
Pulmonary HT

Parasternal Thrill

IN UK grade
AS/MS

Left 2nd ICS - VSD

Heel/Ulnar border placed over left 4th ICS in expiration

Character of apical impulse

Thumping heave → Force of contraction but sustained compared AS, HOCM

Hyperdynamic

MR, AR, VSD

Grading of Parasternal heave

① Visible, palpable

② Visible, palpable, obliterated

③ Visible, palpable, not obliterated

—

11

07/11/17

we can be felt

2-strikes chest wall.

The Doctor and the

ing preinfused

PRESSURE OVERLOAD

Undisplaced Apical impulse

But abnormal forceful
apical impulse,

sd not d

(3174704) SA
L.V. ngprabhu
SA

Compensated A.S.
(Karyomeloidin)

MS (Char-fanning)

AR MR

$\mathbb{D}CM \rightarrow \text{eccentric myocardiium}$

$\text{HO}(\text{M}) \rightarrow \text{concentric myocardiitis}$

As

A

$\mathbb{D}CM \rightarrow \text{eccentric myocardiium}$

$\text{HO}(\text{M}) \rightarrow \text{concentric myocardiitis}$

A

Scanned by CamScanner

Perussion

From periphery towards heart

11d to heart borders

Cardiac border - beyond Mid clavicular line $\rightarrow$ Cardiomegaly

Auscultation

Mitral area

S_1, S_2 heard/not

Intensity of S_1

Added sounds

Grading

Murmur / Systolic / diastolic Timing early / mid / late / pan

Intensity, pitch

Character, radiation, respiration, position

Turbid area

S_1, S_2 heard/not

Intensity of S_1

Added sounds

Grading

Murmur - systolic / diastolic Timing early / mid / late / pan

Intensity
character

Pitch

Radiation

Respiration

Position — Left lateral / forward / bending

Phonogram
 S₁, S₂
 Always all
 synchronous equally
 low freq. sounds
 so audible
 A, P, T, M

Bell
 S₃-S₄
 Transmits all sounds well
 but high freq. murmurs
 not - low freq. sounds
 less audible
 M, T

Left axilla → Parastolic murmur
 of AR (radiation)

Carotid artery → carotid bruit
 ejection systolic murmur
 of MS.

Carotid pulse & S₁ synchronous

Internal B/w S₁ & S₂ → SYSTOLIC

Internal b/w S₂ & S₁ → DIASTOLIC

BEST HEARD SITES

(M) → S₁
 S₃ S₄
 mid diastolic murmur of MS

(T) → Opening snap of MS
 early diastolic murmur of AR, TR
 parastolic murmur of VSD

(P) - S₂
 Pulmonary valve murmurs

(A) - Ejection systolic murmur of AS
 HOCM.

Below left clavicle - Continuous
 machinery murmur
 of persistent PDA

Pulmonary area

S₁ S₂ heard/not

Intensity of S₂

Added sounds

Grading →

Murmur - systolic / diastolic Early / mid / late / per

Timing

- Intensity

- Character

- Pitch

- Radiation

- Respiration

- Position - Left lateral / forward bending

Aortic area

S₁ S₂ heard/not

Intensity of S₂

Added sounds

Grading →

Murmur - systolic / diastolic Early / mid / late / per

Timing

- Intensity

- Character

- Pitch

- Radiation

- Respiration

- Position - Left lateral / forward bending

DATE

Other system examinations

RS

NVBS

Basal crepitation

Abdomen

Any added sounds

Organomegaly

Fluid thrill

CNS

NFND

Diagnosis

Investigations

Treatment